

Determining the C_v and Heat Capacity Ratio for N_2 , He, and CO_2

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PURPOSE

The purpose of this investigation is to determine the heat capacity at constant volume (C_v) of three gases, Helium (He), Carbon Dioxide (CO_2), and Nitrogen (N_2) under isochoric conditions.

This was achieved by determining the heat capacity ratio, $\gamma = \frac{C_p}{C_v}$, through the use of measurements obtained from the adiabatic expansion of each gas. These experimentally determined C_v values were then compared to values predicted by statistical thermodynamics and literature values. The changes in pressure during the adiabatic expansion were used to calculate both the C_v and γ for each of the gases.

THEORY

From the First Law of Thermodynamics, Mayer's relationship can be derived, which states that the difference between the molar heat capacity at constant pressure ($C_{p,m}$) and at constant volume ($C_{v,m}$) is equal to the gas constant (R). This can be applied to this experiment to calculate the experimental C_v value.

Equation 1: Mayer's Relationship

$$C_p = C_v + R$$

The heat capacity (C_v) can also be calculated theoretically using statistical thermodynamics, given that the shape of the molecule and its number of degrees of freedom are known. For a

molecule with N atoms, there are 3N total degrees of freedom, consisting of translational, rotational, and vibrational contributions to motion.

Equation 2: Equipartition Theorem

$$C_v = C_{translational} + C_{rotational} + C_{vibrational}$$

These contributions depend on molecular geometry. Translational motion contributes $\frac{3R}{2}$ for all molecules. Rotational motion contributes R for linear molecules and $\frac{3R}{2}$ for non-linear molecules. Vibrational modes generally contribute only at higher temperatures; the number of vibrational modes is $3N - 5$ for linear and $3N - 6$ for non-linear molecules. The vibrational contribution to the C_v from each mode is calculated using the Einstein expression:

Equation 3: Einstein Vibrational Heat Capacity Equation

$$C_{V,Vibrational} = R \sum \left(\frac{x_i^2 e^{x_i}}{(e^{x_i} - 1)^2} \right)$$

Where x_i is equal to $\frac{hcv_i}{k_b T}$, with h as Planck's constant, c the speed of light, k_b Boltzmann's constant, and v_i the vibrational wavenumber of mode i.

Since no heat is exchanged in this experiment, the First Law reduces to $\Delta U = w$, since $q = 0$ for an adiabatic expansion. After applying the ideal gas law and integrating, the heat capacity ratio (γ) equation can be expressed as:

Equation 4: Heat Capacity Ratio

$$\gamma = \frac{\ln(P_1/P_2)}{\ln(P_1/P_3)}$$

In this equation, P_1 is the initial pressure, P_2 is the atmospheric pressure, and P_3 is the final equilibrated pressure after expansion and re-stoppering. Finally C_v can be calculated from:

Equation 5: Relation Between C_v and γ

$$C_v = \frac{R}{(\gamma - 1)}$$

EXPERIMENTAL

Equipment: Large airtight carboy, rubber stopper, compressed gas tanks (He , N_2 , CO_2) with regulators, α -bromonaphthalene oil manometer, standard manometer, stopcock, and supply hose.

Experiment:

The gas supply hose was connected to the carboy, and the system was flushed with the selected gas for approximately 3 minutes to displace air from the system. The stopper was held in place while the exhaust stopcock was closed, pressurizing the carboy until the oil manometer column difference reached 600-700mm of oil. The needle valve and stopcock were then closed, sealing the system and allowing it to equilibrate for 5 minutes until the column heights were stable. The initial pressure difference, ΔP_1 (mm oil), was recorded.

The stopper was removed, allowing the gas to expand for approximately 1 second, then immediately replaced securely. The system was allowed to re-equilibrate for 5 minutes, after which the final pressure difference, ΔP_3 (mm oil) was recorded. The procedure was repeated twice for each gas. Atmospheric pressure P_2 was measured using the barometer for each set of trials.

The Oil manometer readings needed to be converted from mm of oil to mm of Hg using the density ratio of α -bromonaphthalene (1.48g/mL) to mercury (13.6 g/mL):

Equation 6: Oil to Mercury Conversion

$$P_i = P_2 + \Delta P_{(mm\ oil)} * \left(\frac{1.48g/mL}{13.6g/mL} \right)$$

DATA

The average atmospheric pressure for the trials was measured as 762.9 mm Hg. The absolute pressures P_1 and P_3 were calculated from the oil manometer readings using the conversion for α -bromonaphthalene to mercury ($\frac{1.48}{13.6}$). The heat capacity ratios (γ) were then determined, and

the isochoric heat capacity, C_v , was calculated for each trial using **Equations 4 and 5**. The results are reported in Tables 1-3 below.

Table 1: Carbon Dioxide (CO_2)

Trial	ΔP_1 (mm oil)	P_1 (mm Hg)	ΔP_3 (mm oil)	P_3 (mm Hg)	γ	C_v ($\frac{J}{mol \cdot K}$)
1	343.0	800.226	77.1	771.290	1.2970	27.99
2	373.5	803.546	70.0	770.518	1.2367	35.12
Avg	-	-	-	-	1.2669	31.56

Table 1: Measured and calculated values for carbon dioxide. $p_2 = 762.9 \text{ mm Hg}$

Table 2: Helium (He)

Trial	ΔP_1 (mm oil)	P_1 (mm Hg)	ΔP_3 (mm oil)	P_3 (mm Hg)	γ	C_v ($\frac{J}{mol \cdot K}$)
1	348.0	800.771	97.0	773.456	1.3959	21.00
2	371.2	803.295	116.2	775.545	1.4676	17.78
Avg	-	-	-	-	1.4318	19.39

Table 2: Measured and calculated values for helium. $p_2 = 762.9 \text{ mm Hg}$

Table 3: Nitrogen (N_2)

Trial	ΔP_1 (mm oil)	P_1 (mm Hg)	ΔP_3 (mm oil)	P_3 (mm Hg)	γ	C_v ($\frac{J}{mol \cdot K}$)
1	347.5	800.716	103.8	774.196	1.4364	19.05
2	367.0	802.838	96.0	773.347	1.3634	22.88
Avg	-	-	-	-	1.3999	20.96

Table 3: Measured and calculated values for nitrogen. $p_2 = 762.9 \text{ mm Hg}$

CALCULATIONS

Example Calculations: CO_2 Trial 1

Convert oil manometer readings to absolute pressure (mm Hg) (**Equation 6**):

$$P_1 = P_2 + \Delta P_1 * \left(\frac{1.48}{13.6}\right)$$

$$P_1 = 762.9 + 343.0 * \left(\frac{1.48}{13.6}\right) = 800.226 \text{ mm Hg}$$

$$P_3 = 762.9 + 77.1 * \left(\frac{1.48}{13.6}\right) = 771.290 \text{ mm Hg}$$

Calculate the heat capacity ratio (γ) (**Equation 4**):

$$\gamma = \frac{\ln(P_1/P_2)}{\ln(P_1/P_3)}$$

$$\gamma = \frac{\ln\left(\frac{800.226}{762.9}\right)}{\ln\left(\frac{800.226}{771.290}\right)} = 1.2970$$

Calculate the C_v from Mayer's relation and the definition of γ (**Equation 5**):

$$C_v = \frac{R}{(\gamma-1)}$$

$$C_v = \frac{8.314}{(1.2970-1)} = 27.99 \text{ J/mol} * K$$

Error Propagation:

Table 4: Final Error Propagation Values

Gas	Trial Number	Error from P_1	Error from P_3
CO_2	1	0.001008	0.004566
CO_2	2	0.000702	0.003824
He	1	0.001425	0.005200
He	2	0.001656	0.005383
N_2	1	0.001618	0.005508
N_2	2	0.001210	0.004711

Table 4: Propagated error for each trial (See Supplemental Information for Maxima)

RESULTS

The experimental, theoretical, and literature values of the C_v for all three gases are reported in Table 5. The literature values are the $C_p - R$ values pulled from the NIST database. The average experimental γ and percent error relative to the literature values are also reported.

Table 5: Experimental, Theoretical, and Literature Values

Gas	Experimental C_v (J/mol*K)	Theoretical C_v (J/mol*K)	Literature C_v (J/mol*K)	% Error	Experimental γ
CO_2	31.56	28.74	28.81	9.5%	1.2669
He	19.39	12.47	12.48	55.4%	1.4318
N_2	20.96	20.80	20.76	1.0%	1.3999

Table 5: Comparison of experimental, theoretical, and literature values for He, N_2 , and CO_2 .

Question 1 - Comparison of Experimental C_v with Theoretical and Literature Values

Nitrogen produced the best agreement with the theoretical and literature values, with an experimental value of 20.96 J/mol*K compared to the literature value of 20.76 J/mol*K and the theoretical value of 20.80 J/mol*K, yielding a final percent error of 1.0% when compared to the literature value. Nitrogen is a linear diatomic molecule, and it has a singular high-frequency vibrational mode that is largely inactive at room temperature. This explains why the experimental value from this investigation aligns closely with the theoretical and literature C_v values. Carbon dioxide showed a moderate deviation with an experimental value of 31.56 J/mol*K compared to the theoretical value of 28.74 J/mol*K and the literature value of 28.81 J/mol*K. CO_2 is a linear triatomic molecule with four vibrational modes, including two that are partially active at room temperature. Helium showed the largest deviation, with an experimental value of 19.39 J/mol*K compared to a literature value of 12.48 J/mol*K. Helium is a small monatomic noble gas that only has translational degrees of freedom. The large overestimate is most likely due to the high sensitivity of γ to small pressure differences, which is amplified for monatomic gases like helium.

Question 2 - Effect of Re-stoppering too quickly

If the carboy is re-stoppered too quickly after removal, the gas inside would not have sufficient time to expand to atmospheric pressure. The internal pressure would remain above P_2 , meaning that the P_3 value would be higher than expected. This decreases the denominator of the γ expression, increasing γ , which would decrease the calculated C_v value.

Question 3 -Effects of Re-stoppering too slowly

If the carboy is re-stoppered too slowly after removal, the gas expands to atmospheric pressure but begins to warm before the system is sealed. As the gas warms, its pressure increases above the true post-expansion value. As a result of this, the measured P_3 value will be higher than the true value. This would reduce the denominator in the γ expression, increasing the value of γ . This would decrease the calculated C_v value. In reality, restoppering too slowly or too quickly can influence the variability in either direction.

Question 4 - Percent Error

The percent error for this experiment was reported above in Table 5. The percent errors were taken relative to the literature values and were calculated to be 9.5% for CO_2 , 55.4% for He, and 1.0% for N_2 . Nitrogen was the smallest error and therefore the most reliable result from the experiment. The large error reported for the helium value can likely be attributed to helium being a fast, light gas that would be more sensitive to the calculations from the gamma equation for a monatomic gas. The error for CO_2 can likely be attributed to inconsistent stopper timing, as there is a large spread between Trial 1 (27.99 J/mol*K) and Trial 2 (35.12 J/mol*K).

Question 5 - Origins of Error from Propagation

The propagated error shows that the dominant contribution to the uncertainty in γ is from P_3 , as the uncertainty is approximately 3 to 4 times larger than the error from P_1 . An example of this can be found with CO_2 trial 1, in which the error from P_1 was 0.001008 while the error from P_3

was 0.004566. This pattern is fairly consistent in all trials for all gases. This is likely expected as P_1 is measured after a 5-minute equilibration period, which would produce stable conditions, and P_3 depends on the timing of the stopper replacement. It is also possible that instrumentation contributed to the deviation from the literature value, but most of the error is likely attributed to stopper timing rather than instrumental precision.

REFERENCES

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SUPPLEMENTAL INFORMATION

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(%i13) sigmaP1: 0.1·torr;
sigmaP1 0.1 torr
(%i14) sigmaP3: 0.1·torr;
sigmaP3 0.1 torr
(%i15) P2: 762.9·torr;
P2 762.9 torr
(%i16) P1exp: [800.226, 803.546, 800.771, 803.295, 800.716, 802.838]·torr;
P1exp [800.226 torr, 803.546 torr, 800.771 torr, 803.295 torr, 800.716 torr, 802.838 torr]
(%i17) P3exp: [771.290, 770.518, 773.456, 775.545, 774.196, 773.347]·torr;
P3exp [771.29 torr, 770.518 torr, 773.456 torr, 775.545 torr, 774.196 torr, 773.347 torr]
(%i18) Nexp: length(P1exp);
Nexp 6
(%i19) v(P1,P3) := log(P1/P2) / log(P1/P3);
(%o19) 
$$v(P1, P3) := \frac{\log\left(\frac{P1}{P2}\right)}{\log\left(\frac{P1}{P3}\right)}$$

(%i20) dvdP1(P1,P3) := diff(v(P1,P3), P1, 1);
(%o20) 
$$\text{dvdP1}(P1, P3) := -\frac{d}{d P1} v(P1, P3)$$

(%i21) dvdP3(P1,P3) := diff(v(P1,P3), P3, 1);
(%o21) 
$$\text{dvdP3}(P1, P3) := -\frac{d}{d P3} v(P1, P3)$$

(%i22) ErrorinP1(P1,P3) := sqrt(dvdP1(P1,P3)^2 · sigmaP1^2);
(%o22) 
$$\text{ErrorinP1}(P1, P3) := \sqrt{\text{dvdP1}(P1, P3)^2 \cdot \text{sigmaP1}^2}$$

(%i23) ErrorinP3(P1,P3) := sqrt(dvdP3(P1,P3)^2 · sigmaP3^2);
(%o23) 
$$\text{ErrorinP3}(P1, P3) := \sqrt{\text{dvdP3}(P1, P3)^2 \cdot \text{sigmaP3}^2}$$

(%i24) for i:1 thru Nexp do disp(at(ErrorinP1(P1,P3), [P1=P1exp[i], P3=P3exp[i]]));
0.0010076435696405942
7.019335010049269 10-4 0.0014246925272217833 0.0016557790836619207
0.001618104050361075 0.0012095207806943073
(%o24) done
(%i25) for i:1 thru Nexp do disp(at(ErrorinP3(P1,P3), [P1=P1exp[i], P3=P3exp[i]]));
0.0045657779239251855 0.0038241931862076904 0.005200269407656054 0.005382721640968295
0.005508491689467517 0.004710760704561866
(%o25) done

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